

SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /
5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems.*
(BL6)

Assessment Tool Number	: CO3_AT1
Assessment Tool Weightage	: 35%
Assessment Tool	: Design Task
Duration	: 180 Minutes
Marks	: 100 Marks
Type of Assessment Conduction	: At Home
Evaluation Type	: GitHub
Result Generation	: Students design and justify appropriate wireless communication solutions by applying engineering principles to meet specified requirements.
Topic Covered	: Spectrum management, licensed/unlicensed spectrum, dynamic spectrum access, cognitive radio, OFDMA, resource block allocation, GFDMA, NOMA, power-domain multiplexing, SIC, and IoT applications. Advanced waveforms and wireless technologies including GFDM, FBMC, UFMC, mmWave propagation, path loss and penetration, millimeter-wave antenna systems, MIMO, and Massive MIMO fundamentals.

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Code of Conduct:

*I JANANI.S.R(192512090) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

13. Design a centralized traffic control model using Cloud RAN principles.

CODE:

clc;

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clear;
close all;

%% Centralized Traffic Control using Cloud RAN
% Simulation Parameters

numJunctions = 5;      % Number of traffic junctions
simulationTime = 100;   % Simulation time (s)
roadLength = 1000;     % Road length (m)

numVehicles = 100:100:1000;
trafficDensity = numVehicles / roadLength;

bandwidth = 20e6;      % 20 MHz
Pt = 23;               % Transmit power (dBm)
SNR_dB = 0:2:20;       % SNR range

%% Junction Deployment

x = 100:200:900;
y = 500*ones(1,numJunctions);

figure;
scatter(x,y,100,'filled');
grid on;
xlabel('Road Position (m)');
ylabel('Road Width (m)');
title('Traffic Junction Deployment');
hold on;

for i = 1:numJunctions
    text(x(i),y(i)+40,['Junction ',num2str(i)]);
end

%% Traffic Density Analysis

vehicleCount = 100:100:1000;
trafficDensity = vehicleCount / roadLength;

figure;
plot(vehicleCount,trafficDensity,'-o','LineWidth',2);
grid on;
xlabel('Number of Vehicles');
ylabel('Traffic Density (vehicles/m)');
title('Traffic Density vs Number of Vehicles');

%% Cloud RAN Processing Load

processingLoad = 20 + 0.07*vehicleCount;

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figure;
plot(vehicleCount,processingLoad,'-s','LineWidth',2);
grid on;
xlabel('Number of Vehicles');
ylabel('Cloud Processing Load (%)');
title('Cloud RAN Processing Load vs Traffic');

%% Throughput Analysis

time = 1:simulationTime;

throughput = 80 + 10*sin(time/10) + 5*rand(1,simulationTime);

figure;
plot(time,throughput,'LineWidth',2);
grid on;
xlabel('Time (s)');
ylabel('Throughput (Mbps)');
title('Cloud RAN Throughput vs Time');

%% Latency Analysis

latency = 5 + 0.015*vehicleCount;

figure;
plot(vehicleCount,latency,'-d','LineWidth',2);
grid on;
xlabel('Number of Vehicles');
ylabel('Latency (ms)');
title('Latency vs Number of Vehicles');

%% BER Calculation

BER = 0.5*erfc(sqrt(10.^(SNR_dB/10)));

figure;
semilogy(SNR_dB,BER,'-o','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate');
title('BER vs SNR');

%% Traffic Signal Optimization

normalGreenTime = 30*ones(1,length(vehicleCount));

optimizedGreenTime = 30 + ...
    20*(vehicleCount-min(vehicleCount)) / ...
    (max(vehicleCount)-min(vehicleCount));

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figure;
plot(vehicleCount,normalGreenTime,'--','LineWidth',2);
hold on;
plot(vehicleCount,optimizedGreenTime,'-o','LineWidth',2);
grid on;

xlabel('Number of Vehicles');
ylabel('Green Signal Time (s)');
title('Traffic Signal Optimization using Cloud RAN');

legend('Normal Control','Centralized C-RAN Control');

%% Packet Delivery Ratio

packetLoss = 2 + 0.008*vehicleCount;
PDR = 100 - packetLoss;

figure;
plot(vehicleCount,PDR,'-s','LineWidth',2);
grid on;
xlabel('Number of Vehicles');
ylabel('Packet Delivery Ratio (%)');
title('Packet Delivery Ratio vs Traffic Load');

%% Resource Utilization

resourceUtilization = 40 + ...
    0.05*vehicleCount;

figure;
plot(vehicleCount,resourceUtilization,'-^','LineWidth',2);
grid on;
xlabel('Number of Vehicles');
ylabel('Resource Utilization (%)');
title('Cloud RAN Resource Utilization');

%% Performance Summary

avgThroughput = mean(throughput);
avgLatency = mean(latency);
avgBER = mean(BER);
avgPDR = mean(PDR);
avgResource = mean(resourceUtilization);

fprintf('\n===== Centralized Traffic Control using Cloud RAN =====\n');

fprintf('Number of Junctions      : %d\n',numJunctions);
fprintf('Maximum Number of Vehicles: %d\n',max(vehicleCount));
fprintf('Average Throughput          : %.2f Mbps\n',avgThroughput);
fprintf('Average Latency              : %.2f ms\n',avgLatency);

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fprintf('Average BER          : %.5f\n',avgBER);
fprintf('Average Packet Delivery : %.2f %%\n',avgPDR);
fprintf('Average Resource Usage    : %.2f %%\n',avgResource);
fprintf('Optimized Green Time      : %.2f s\n',mean(optimizedGreenTime));

```

OUTPUT:



